# Introduction to Data Analytics
# MS - 4610

Session 13 – Bootstrapping, Bagging, Random Forests and Boosting

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# Bootstrapping

- A non-parametric way to quantify the uncertainty in an estimate
  - Estimate of $\bar{X}$
  - Co-efficient estimates in Regression
    - X – random
    - X – fixed
  - How does it broadly apply as a technique related to Bias-Variance dichotomy

# Bootstrapping

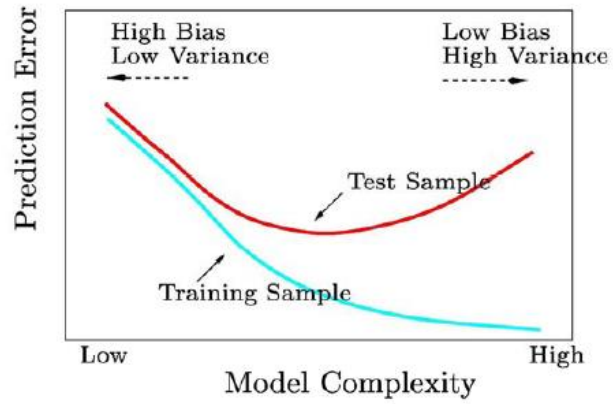
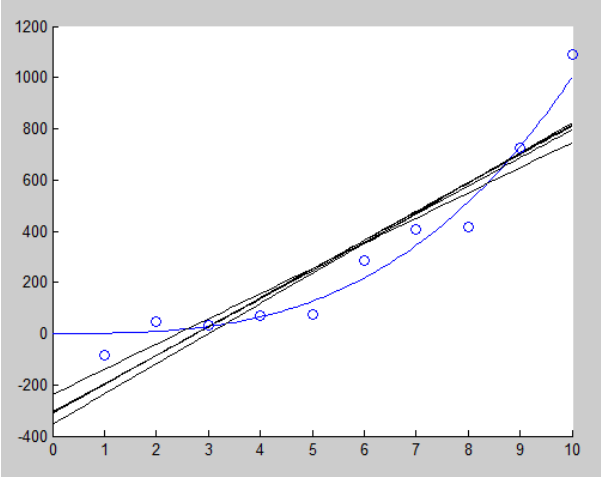


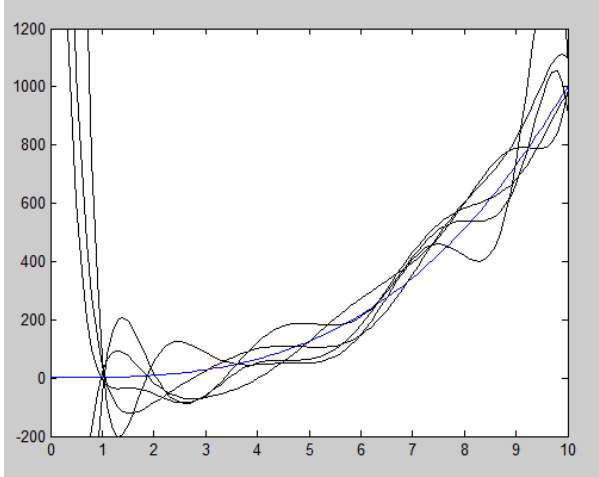

Figure 7.1: *Behavior of test sample and training sample error as the model complexity is varied.*

- Quantifies variance in the model
- Basic proofs on selection of points
- $1 - (\frac{n-1}{n})^n$

# Bagging

- Bagging and Boosting are called Ensemble techniques or additive methods
- Bagging: Bootstrap aggregating is a general purpose approach for reducing the variance of statistical learning methods
- Remember:



# Bagging

- Problem is that we don't have multiple training sets (it was a thought experiment)
- So what, let's bootstrap!
- Works well with greedy algorithms like Trees and complex models

# Random Forests

- Can we just bag Trees and get great prediction error?
- Random forests further decorrelates trees by picking a subset of predictors (called random subspace method). Rule of thumb is to pick $\sqrt{k}$ predictors.
- Another technique is random split selection

# Boosting

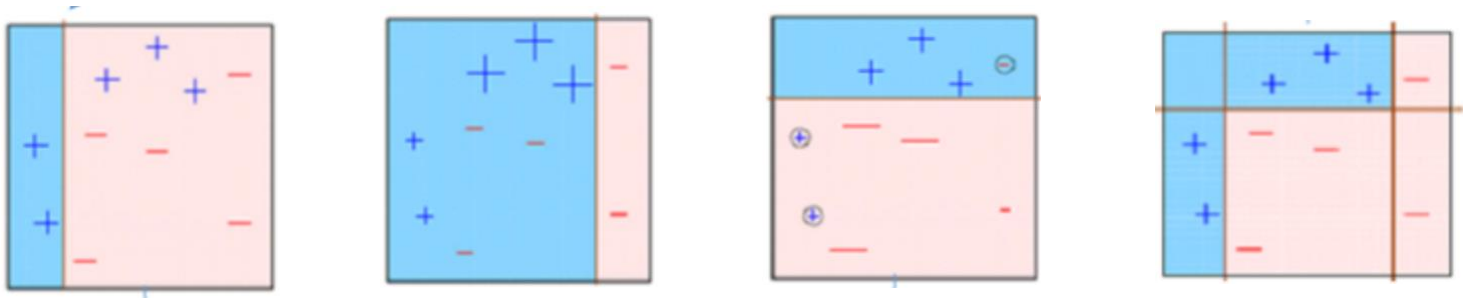
- Also an additive/ensemble method like bagging where multiple versions of a weak classifier are built.
- However, Boosting builds models sequentially (and therefore not independent)

# Boosting

- Algorithm from ISLR

**Algorithm 8.2** *Boosting for Regression Trees*

1. Set $\hat{f}(x) = 0$ and $r_i = y_i$ for all $i$ in the training set.
2. For $b = 1, 2, \ldots, B$, repeat:
   (a) Fit a tree $\hat{f}^b$ with $d$ splits ($d+1$ terminal nodes) to the training data $(X, r)$.
   (b) Update $\hat{f}$ by adding in a shrunken version of the new tree:
   $$\hat{f}(x) \leftarrow \hat{f}(x) + \lambda \hat{f}^b(x). \tag{8.10}$$
   (c) Update the residuals,
   $$r_i \leftarrow r_i - \lambda \hat{f}^b(x_i). \tag{8.11}$$
3. Output the boosted model,
   $$\hat{f}(x) = \sum_{b=1}^{B} \lambda \hat{f}^b(x). \tag{8.12}$$

# Example with Classification

- Example from [https://medium.com/greyatom/a-quick-guide-to-boosting-in-ml-acf7c1585cb5](https://medium.com/greyatom/a-quick-guide-to-boosting-in-ml-acf7c1585cb5)